51 Additions per Recursive Step for Rank-23 3×3 Matrix Multiplication, Machine-Checked over Arbitrary Rings

Greg Sidebottom

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Abstract

We present a rank-23 bilinear scheme for 3×3 matrix multiplication that uses 51 additions per recursive step in the alternative-basis (Karstadt–Schwartz) cost model, one below Nielsen's 52 (arXiv:2609.06588). The count 51 itself was first posted, on a different scheme, by Stapleton and Perminov (arXiv:2609.32388, September 26, 2026), found by heuristic sparse-basis search; the present work is independent and complementary: exactness and breadth rather than a single construction. The scheme is valid over every ring — the products keep their left/right factor order, so it recurses on block matrices — and its correctness is machine-checked in Lean 4/Mathlib down to the kernel (`Ring51.lean`; the core theorem depends on the axiom `propext` alone). The count is exactly optimal for the scheme's equivalence class: we introduce an exact free-basis minimization procedure whose negative answers are exhaustion proofs, and use it to show that 51 is the minimum for this class, that 52 is exactly optimal both for Nielsen's class and for the class of the current flat-model record, and to sweep the complete Heule–Kauers–Seidl database of rank-23 classes. The sweep — made variant-aware after a name-collision blind spot surfaced — is complete: exactly four classes attain 51, and all four lie in a single HKS invariant family, whose rank-spectrum signature we decode and analyze. We machine-check Stapleton and Perminov's scheme as well, so every ring-valid 51-package with known conversions is now kernel-checked. Separately, we document a systematic gap phenomenon between $\mathbb{F}_2$ and signed optima: the $\mathbb{F}_2$-optimal chains of a class typically admit no sign assignment under most of the class's integer sign models, and certificate lifting fails where a direct signed search succeeds.

1 Introduction

A rank-r bilinear scheme for 3×3 matrix multiplication computes the 9 entries of $C = AB$ using r products of linear combinations of the entries of A and of B. Laderman's $r = 23$ scheme [1] remains the smallest known rank for the 3×3 format ($19 \leq R(\langle 3,3,3 \rangle) \leq 23$ over arbitrary fields [2]; over $\mathbb{F}_2$ the lower bound 20 was shown by Wang [3], and two September 2026 preprints claim 21 [4, 5]). With the rank fixed at 23, attention has turned to the *additive* cost of such schemes, in two distinct models.

The flat model. Count every binary addition or subtraction of a standalone straight-line program (negation free, no preprocessing). The published record chain runs Stapleton 60 [6] → Mårtensson–Wagner–Stapleton 59 [7] → Perminov 58 [8] → Sun 56 [9] → 55 (the author, Zenodo deposit of July 7, 2026 [10]; independently rediscovered by Karunaratne and Idamekorala on arXiv on July 28 [11] — same tensor, same 13+14+28 split, same transposed output construction).

The alternative-basis model. Karstadt and Schwartz [12] observed that a scheme may be conjugated by invertible linear *basis changes* on the two inputs and the output. Used recursively, the basis changes are applied once per matrix, blockwise, at total cost $O(n^2 \log n)$ — asymptotically free — while the additions *inside* the changed basis are paid at every recursive step and multiply the leading constant of the $O(n^{\log_3 23})$ term. Their flagship result reduced Strassen's 2×2 step from 15 to 12 additions. For rank-23 3×3, sparse-decomposition work of Beniamini and Schwartz [13] gave 61 per step; Nielsen [14] gave 52, searching random basis changes over $\mathbb{F}_2$ and lifting to general fields; and Stapleton and Perminov [15] gave 51 on a flip-graph-derived scheme, with remarkably sparse conversions (five additions in total, so their complete one-shot algorithm costs 56 in ordinary coordinates). We prove below that their kernel, too, is exactly optimal for its class, and that their one-shot 56 is optimal for the class as well (its exact flat minimum). The two models price different objects and are not comparable entry-by-entry: a flat scheme is an alternative-basis scheme with identity bases, so the alternative-basis optimum can only be smaller, and conversely a per-step count omits the (amortized) basis-change additions that the flat model charges.

Contributions.

1. **Three further 51-addition-per-step classes** (Section 4), found by exact search rather than sampling and pairwise inequivalent to each other and to the scheme of Stapleton and Perminov [15] — so at least four distinct classes attain 51. Two of the three are realized over every ring: `i12w211c23fg` (sign model 21, conversions 11, Lean-checked) and `i12w227c23ci-012` (sign model 15); the third, `i12w219c23fg`, so far stands over $\mathbb{F}_2$ only (Section 7). All four classes lie in a single HKS invariant family (Section 6).

2. **Exact per-class optimality** (Section 3): an exact *free-basis minimization* procedure for one leg of a scheme, with machine-checkable certificates for upper bounds and finite exhaustions for lower bounds. The three legs decouple, so per-class per-step optima are exact: 51 is optimal for its class; 52 is exactly optimal for Nielsen's class (his published count, now with a matching lower bound) and for the class `i19w225c4efh` of the flat-model record (down from 55 flat).

3. **A database-wide sweep** (Section 5): a basis-invariant counting floor for every scheme of the HKS database of 17,376 rank-23 schemes, and a complete exact sweep of all 561 stored variants whose floor permits ≤ 51 — run variant-aware after a name-collision blind spot surfaced, the completion that recovered Stapleton and Perminov's class and found the fourth. The database's ≤ 51 census is final.

4. **The $\mathbb{F}_2$–$\mathbb{Z}$ sign gap** (Section 7): the $\mathbb{F}_2$ optimum of a leg is often unattainable over $\mathbb{Z}$ under most of the class's sign models. For the 51-class, 20 of the first 22 sign models admit no signed chain matching the $\mathbb{F}_2$ counts, and certificate lifting (the standard route, used by Nielsen [14]) fails on 900 sampled certificates, 64 diagonal gauges and 128 models — while a direct signed search succeeds, instantly, at sign model 21.

5. **End-to-end machine verification** (Section 8): search certificates with independent replay; a word-ring verification of the assembled program over $\mathbb{Z}\langle A, B\rangle$; and a Lean 4 proof over arbitrary (non-commutative) rings, reproducible in seconds (Appendix A). We extend the same treatment to Stapleton and Perminov's published program (`proof/Mm/SP51.lean`), so both ring-valid 51-packages with known conversions are kernel-checked.

2 The cost model

A rank-23 scheme is a triple (U, V, W) of 23×9 integer matrices with $C[p,q] = \sum_{m=1}^{23} W_{m,(p,q)} (U_m \cdot \operatorname{vec} A)(V_m \cdot \operatorname{vec} B)$. Here vec flattens a 3×3 matrix row-major into a 9-vector (coordinate k is entry $(\lfloor k/3 \rfloor, k \bmod 3)$). In the alternative-basis model one chooses invertible 9×9 matrices B_U, B_V, B_W (unimodular over $\mathbb{Z}$ for ring validity), computes $A' = B_U^{-1} \operatorname{vec} A$ and $B' = B_V^{-1} \operatorname{vec} B$ once per matrix, runs the recursion entirely in the primed coordinates, and decodes the output once. The *per-step cost* is the number of additions needed to evaluate the three in-basis linear maps: the two input legs UB_U, VB_V (each 23 forms from 9 inputs) and the output leg (the 9×23 map $m \mapsto (WB_W^{\top})^{\top} m$, priced through the transposition principle).

Proposition 2.1 (transposition overhead)**.** *For a 9×23 output map with no zero rows or columns, the minimal addition count equals that of its 23×9 transpose plus $23 - 9 = 14$.*

This is Tellegen's principle, used constructively throughout (the flat record's 28-addition output is a transposed 14-addition chain [10, 11]; the 51 scheme's 26-addition output is a transposed 12-addition chain). Hence a per-step total is $a_A + a_B + (a_C + 14)$ where a_A, a_B, a_C are the three 23×9 leg costs, and 51 corresponds to legs $(12, 13, 12)$.

Lemma 2.2 (leg decoupling over rings)**.** *Over any ring, with negation free, the per-step optimum of a class is the sum of the three leg optima, and the per-product sign constraint $\varepsilon_u \varepsilon_v \varepsilon_w = +1$ is vacuous: leg cost is invariant under per-row sign flips, and whatever signs the three chains realize are absorbed at the point of use (a -1 coefficient in the output assembly is free).*

Proof. A chain value used with a minus sign costs nothing extra in the negation-free model, so replacing any target row by its negation changes no leg cost. Given chains realizing $\varepsilon_u^{(m)} u_m$, $\varepsilon_v^{(m)} v_m$, $\varepsilon_w^{(m)} w_m$, use the output value of product m with coefficient $\varepsilon_u^{(m)} \varepsilon_v^{(m)} \varepsilon_w^{(m)}$; the composite computes $\sum_m (u_m A)(v_m B)\, w_m$ exactly. □

3 Exact free-basis minimization

Fix one leg: a 23×9 matrix X over $\mathbb{F}_2$ (or over $\mathbb{Z}$, with targets taken up to sign) with d distinct nonzero rows. The problem is to minimize, over invertible B, the number of additions needed to compute all rows of XB from the 9 transformed inputs. Equivalently: choose any 9 linearly independent *free* forms, then build every distinct target by pairwise additions.

Proposition 3.1 (structure of optimal chains)**.** *A chain with a additions, k of whose 9 free forms are targets, and h helper (non-target) additions, satisfies $a = (d - k) + h$. Consequently $a = d - 9$ iff some independent 9-target basis closes to all of T under the waste-free closure (repeatedly add any pair-sum that is an unbuilt target); $a = d - 8$ iff a 9-target basis plus one helper, injectable at the closure fixpoint, suffices, or an independent 8-target basis plus one arbitrary free vector; and so on.*

Two facts make this an exact procedure. The waste-free closure is *monotone* — available forms never hurt — so greedy closure decides waste-free reachability. And every form preceding the i-th helper lies in the pure closure of the basis and the earlier helpers, so helpers are enumerable at successive fixpoints: the case sweep is finite and complete for the 9-target-basis configurations. (The tiers with free non-target basis vectors are finite over $\mathbb{F}_2$ — 511 vectors — and were swept there; over $\mathbb{Z}$ they are unbounded and are the one stated scope caveat of the signed lower bounds; see Section 7.) Every positive answer emits the chain, which an independent checker replays (basis

independence and, over $\mathbb{Z}$, unimodularity by exact determinant; each addition; coverage) before it is reported; every negative answer is a finite exhaustion.

Proposition 3.2 (basis-invariant floor)**.** *In every basis, an input leg needs at least $d-9$ additions (d = distinct rows; invariant under invertible column action), and the output leg at least $(d-9)+14$. Hence the per-step total of a class is at least $d_A + d_B + d_C - 13$.*

4 The 51-addition scheme

Running the procedure on the class `i12w211c23fg` (HKS family `17x2+52x+4y6+5y4+14y3+z32+2z27`) over $\mathbb{F}_2$ gives exact leg minima

$$a_A = 12, \qquad a_B = 13, \qquad a_C = 12,$$

each with a certificate, and with a=11 exhausted on all three legs and a=12 exhausted on the B leg. By Lemma 2.2 the class's per-step optimum is exactly

$$12 + 13 + (12 + 14) \;=\; \mathbf{51}.$$

The ring-valid realization comes from the class's integer sign model 21 (Section 7): signed chains with the same counts, from unimodular signed bases, assembled into the scheme that Appendix A exhibits in full as a Lean definition, together with the kernel-checked theorem that it computes AB over every ring.

The conversions of this realization cost $4 + 4 + 3 = 11$ additions — minimized *exactly for its sign model*, by sweeping every feasible configuration at the optimal leg counts: bases of $k \in \{9, 8, 7\}$ target rows with the remaining slots filled by unit vectors (cost-free rows, with the helper budget $h = a - (d - k)$ adjusted accordingly), scored by $\sum(\text{weight} - 1)$ over the basis rows. Conversion cost depends on the representative (its target rows' raw weights), so we state it per representative. Over the other signings of this representative the minimum is only bracketed. A leg with slack s over its floor needs at least $9-s$ target rows in its basis; those rows must be independent over $\mathbb{F}_2$ (a necessary condition for unimodularity), and row weights do not depend on signs. The lightest such choice gives a sign-independent lower bound of $2+3+2 = 7$, so this representative's optimum lies in $[7, 11]$. Its 4096 sign models fall into 64 gauge orbits. A sweep of all 64 found one further orbit that lifts, also at exactly 11; eight orbits were refuted, and the remaining 54 were undecided within a 600-second budget per leg check, so the sweep is not exhaustive. The complete one-shot algorithm costs $51 + 11 = 62$ additions in ordinary coordinates; for comparison, Stapleton and Perminov's class achieves $51 + 5 = 56$ [15], and the flat-model record is 55 [10, 11]. The same sweep applied to *their* published signing returns conversion costs $1 + 3 + 1 = 5$. Their representative has exactly 128 sign models (exhaustive enumeration), in two gauge orbits; both lift, at conversions 5 and 6. So 5 is the minimum over every $\mathbb{Z}$-lift of their representative within the swept configurations.

A stronger, class-wide statement follows from the flat model. Any one-shot algorithm is itself a flat program, so a class's flat minimum bounds every one of its packages from below. We compute the exact $\mathbb{F}_2$ flat minimum of a class over *all* its representatives. The flat cost is the sum of the three legs' exact addition-chain costs from unit vectors, plus 14 for the output side by transposition. Each leg's cost depends on only two of the three sandwich factors $(P, Q, R) \in \mathrm{GL}_3(\mathbb{F}_2)^3$, so three exact 168×168 tables and a min-plus scan over all 168^3 triples, in each of the six slot variants, cover the whole orbit. $\mathbb{F}_2$ costs bound every $\mathbb{Z}$ signing from below. As a gate, the procedure returns exactly 55 on the flat record's class, a value known from both sides. On Stapleton and Perminov's class it returns 56: no package on their class can cost less than 56 one-shot, so their $51 + 5 = 56$

is exactly optimal for the class, conversions included, and it attains their class's flat optimum. The same computation gives flat floors of 57 for `i12w211c23fg` (so its conversions are at least 6 class-wide, against the 11 found) and for `i12w227c23ci-012`, and 58 for `i12w219c23fg`.

Run over the whole database, it settles the uniqueness of the flat record. A flat 55 needs either input sides of at most 27 (a region we searched exhaustively in [10], where only the record class reaches 55) or an $\mathbb{F}_2$ output side of 27. Exactly 616 database schemes (620 slots of 44,125) admit the latter. On all 616 the exact flat minimum is at least 56, except the record class itself at exactly 55: *the flat-55 record is unique to its class across the database.* Six frontier classes sit at exactly 56, among them Sun's and Stapleton and Perminov's; the latter is Perminov's earlier cn119 class [8] (exact equivalence check). Both attain 56, so both are exactly optimal for their classes. The rest fall between 57 and 63.

Conversion cost at optimal counts is bounded below by the class's light-row supply (the accounting $a = (d - k) + h$ keeps most basis rows among the targets), so near-identity conversions like [15]'s require a class rich in weight-one rows.

The second class, `i12w219c23fg`, has exact legs $(11, 13, 13)$: its A leg *achieves the counting floor* $d - 9$ — a 9-target basis whose waste-free closure covers all 20 targets. Floor-achieving legs are uncommon but not unique (32 of the 1,079 legs decided exactly across the completed sweeps, occurring at $d = 20$, 21 and 22). They share a necessary structure: a floor chain has $d-9$ additions, each producing a target from two targets, so at least $d-9$ of the leg's rows are each the sum of two other rows (rows x, y, z with $x + y = \pm z$). Most observed floor legs also have every row in such a triple, but not all: five of the 32 have an isolated row, which must then be a basis row never used as an operand.

The third class, `i12w227c23ci-012`, was found by the variant-aware completion of the database sweep (Section 5) in the name-collision blind spot: it is the thirteenth stored variant under a name whose first variant is a different de Groote class. Its exact legs are $(14, 12, 11)$ — the *output* leg achieves the counting floor $d - 9 = 11$, the second floor-achieving leg observed and the first on the output side. Direct signed search over its gauge-deduped sign models (Section 7) clears all three legs at model 15: the class is ring-valid at 51, and that sign model's conversions cost exactly $3 + 9 + 5 = 17$, a one-shot total of 68.

For calibration, the same exact procedure gives: for Nielsen's base class `i12w195c23ci`, legs $(12, 12, 14)$ — so his published 52 [14] is *exactly optimal* for his class, a lower bound his sampling-based search could not provide; for the scheme of Stapleton and Perminov [15] — which the sweep identifies as database class `i12w219c23ci-008`, itself hidden in the same name-collision blind spot — legs $(13, 12, 12)$: their heuristically-found 51 kernel is likewise exactly optimal for its class (machine-checked in Lean; Appendix B), which is inequivalent to all three of ours; and for `i19w225c4efh` (the class of the flat-model record), legs $(13, 13, 12)$, i.e. per-step 52, down from 55 flat — with the curious feature that its *output* leg, at 14 flat, is the leg that drops to 12 under basis change.

Table 1 summarizes the four known 51-classes.

5 Database sweep

Computing (d_A, d_B, d_C) for all 17,376 schemes of the HKS database takes seconds and yields the floor distribution of Proposition 3.2: minimum 47, attained by 44 schemes (including two of the four 51-classes, ours and Stapleton and Perminov's; the other two sit at 48); 507 schemes at 48; Laderman alone at 56 ($d = (23, 23, 23)$: no shared factors, maximally unsuited to the model). The exact sweep covers every stored scheme whose floor permits ≤ 51. Stored schemes that share a name (suffixes `-000`, `-001`, …) need not be equivalent: an HKS name is an invariant string plus a hash, and

Table 1: The four known 51-addition-per-step classes (all in HKS family `i12`; Section 6). Legs are the exact free-basis minima (a_A, a_B, a_C); per-step $= a_A + a_B + a_C + 14$; conversion counts are exact for the listed sign model within the swept configuration space (class-wide bounds in the text) (Section 4); one-shot = 51+conversions. "Sign model n" names the n-th SAT-enumerated integer signing of the mask scheme (Section 7) — the first under which a signed chain realizes the legs; the index is enumeration order, for `-012` after quotienting by the 512 diagonal sign gauges.

class	legs	ring-valid	conversions	one-shot	Lean
`i12w211c23fg`	$(12, 13, 12)$	yes (sign model 21)	4+4+3 = 11	62	`Ring51.lean`
`i12w219c23fg`	$(11, 13, 13)^\dagger$	open ($\mathbb{F}_2$ only)	—	—	—
`i12w219c23ci-008` [15]	$(13, 12, 12)$	yes (published)	1+3+1 = 5	**56**	`SP51.lean`
`i12w227c23ci-012`	$(14, 12, 11)^\dagger$	yes (sign model 15)	3+9+5 = 17	68	—

†: contains a floor-achieving leg ($a = d - 9$): the 11-legs, on the input side for `i12w219c23fg` and the output side for `i12w227c23ci-012`.

distinct de Groote classes can share one. A first pass, deduplicated by name, therefore missed some classes, and a second pass evaluated every stored variant individually. The second pass is where Stapleton and Perminov's class (`i12w219c23ci-008`) and the fourth 51-class (`i12w227c23ci-012`) turned up. Together the two passes decide all 561 stored schemes with floor ≤ 48, spanning 23 families:

per-step total	variants	note
51	4	the four classes of Table 1, pairwise inequivalent
52 (tie)	27	
53	56	incl. Stapleton's class at $(13, 13, 13)$
54	39	
55	5	
56	1	
ruled out for ≤ 51	429	leg-sum lower bounds ≥ 38

(One variant, `i110w205c238h-000`, timed out on its third leg; it is nonetheless ruled out rigorously by the floor of the open leg: $15 + {\geq}14 + {\geq}(d_C - 9{=}10) = 39 > 37$. No other leg timed out.) Of the 561 variants, 159 are `i12` and 402 are not: outside `i12`, none reaches 51. We also profiled and swept our own corpus of 123 rank-23 schemes outside the sweep above: 53 walk-discovered classes certified inequivalent to the entire database, plus 70 later finds (not yet equivalence-checked against it). Of these, 16 were eligible at floor ≤ 48, and all 16 are ruled out for ≤ 51 — the database's optimum stands across every class known to us. A 50 would require exact leg sum 36; none exists in the database, at *any* floor. Writing f for the per-step floor, a 50 leaves total slack $50 - f$ across the three legs. At floor ≤ 48 the completed sweep gives minimum sum 38 outside the four 51s. A floor-49 class could host a 50 only with at least two legs floor-achieving (one unit of slack remains), a floor-50 class only with all three, and floor ≥ 51 is arithmetically incompatible. Whether a leg attains its floor is decided exhaustively by the first tier of our procedure: the floor $d - 9$ is attained exactly when some basis of 9 targets closes over all targets. Run on every one of the 2,450 floor-49 and 5,431 floor-50 schemes, it finds *no* floor-49 scheme with two floor-attaining legs and *no* floor-50 scheme with three. The closest approach is slack +2 over the leg floors (the $(11, 13, 13)$ and $(14, 12, 11)$ classes); exact 11-legs exist on both the input and the output side, so sum 36 is not excluded by any per-leg phenomenon observed to date — but any 50 must now come

from outside the published database. Because leg minima only rise with d, the floor-≤ 48 sweep also settles the ≤ 51 question for the entire database.

6 The `i12` family

All four known 51-classes lie in one HKS invariant family. The database groups its 17,376 schemes into 237 families (plus four classically named schemes) by a rank-spectrum string; the family of the four is

`17x2+52x+4y6+5y4+14y3+z32+2z27`,

whose grammar we verified computationally against the data. The x-part is the rank distribution of the 69 coefficient matrices (23 products $\times$ 3 slots): 17 of rank 2 and 52 of rank 1. The y-part is the per-product rank-sum distribution: 4 products of shape $(2, 2, 2)$, 5 summing to 4, and 14 of shape $(1, 1, 1)$. The z-part is the total rank per slot: $(32, 27, 27)$. In the scheme names themselves, `i`N indexes the family (1:1 with the invariant string across the database), `w`N is the scheme's weight as a trilinear form (the number of cubic terms $\sum_m |U_m||V_m||W_m|$ — not its nonzero-entry count), `c`XXXX is a four-character disambiguating hash, and `-`NNN counts stored variants under the same full name — variants that, as Section 5 shows, need not be equivalent.

The concentration is striking: the floor-47/48 region spans dozens of families, but every exact 51 is `i12` — schemes rich in rank-one coefficient matrices (52 of 69) with one heavy slot ($z_A = 32$). The contrast with `i19` — the family of the flat-model record, signature `18x2+51x+4y6+6y4+13y3+z32+z28+z27`, one fewer rank-one matrix — is sharp in the completed sweep data (173 variants decided, zero 51s): its variants repeatedly reach $(13, 13, 12)$-type profiles but stall with one leg proved ≥ 14, yielding ties at 52 and a wall of ruled-outs, never a 51. Whether the rank spectrum *explains* the 51s or merely correlates with them is open. Rank-one-rich legs have many coincident rows, hence small d and dense internal addition triples, which floor chains need (Section 4); the family statistics make this a natural target for a structural lower bound.

7 The sign gap

An $\mathbb{F}_2$ result lifts to the ring-valid statement only if the chains admit consistent signs. A mask scheme fixes only the *support* of the coefficients; an integer *sign model* assigns ± 1 to every nonzero coefficient of (U, V, W) so that each of the 729 Brent equations holds over $\mathbb{Z}$ — spurious terms cancelling in pairs, true terms summing to $+1$ — i.e. it is a genuine integer algorithm whose entrywise absolute value is the stored scheme. A mask typically admits many sign models, and they are genuinely different algorithms: mod 2, $x + y$ and $x - y$ coincide, so leg minima are model-dependent over $\mathbb{Z}$. We enumerate models by SAT with blocking clauses; a model's *index* is just that enumeration order and encodes nothing. Empirically the dependence is drastic:

- **Certificate lifting fails systematically.** For `i19w225c4efh`'s B leg, none of 301 $\mathbb{F}_2$-optimal certificates admits any sign assignment under the database representative's model (256 basis signings each, all unimodular, every one obstructed in an exhaustive depth-first search over the chain's sign choices), nor under all 64 diagonal sign gauges, nor under 128 SAT-enumerated models. The same held for the 51-class's A and B legs.
- **The second 51-class resists.** For `i12w219c23fg`, the first 128 SAT-enumerated sign models — 32 orbits after quotienting by the 512 diagonal gauges, which searches coincide on — all fail the $(11, 13, 13)$ targets; its 51 currently stands over $\mathbb{F}_2$ only.

- **Direct signed search succeeds.** A signed variant of the free-basis procedure (targets up to sign; both signs tried per addition; $|\det|$ checked once per basis, as row signs do not change it) decides each model exactly. For the 51-class, models 0–20 fail (18 cannot reach $a_A = 12$; two reach it and exhaust $a_B = 13$) — and model 21 clears all three legs in seconds, B included. The fourth class repeats the pattern: its models, quotiented by the 512 diagonal sign gauges to 64 orbits, fail until orbit 15 clears $(14, 12, 11)$.
- **The gap is real but model-local.** For `i19w225c4efh`, signed $a_A = 13$ and $a_C = 12$ hold with certificates, but $a_B \geq 14$ over all 9-target-basis configurations: the class's ring-valid per-step value is 52 or 53, with 52 confined to the unbounded free-vector-basis configurations. A bounded version (QF_BV, form entries ≤ 12, basis entries ≤ 2, full-lexicographic symmetry breaking, bit-blasted to CNF) resisted eight solver configurations, the last three being 24-hour runs of CaDiCaL and Kissat on plain and symmetry-broken instances: a documented open question (Section 9). Certificate lifting is thus doubly incomplete: it samples chains, and it misses signed chains whose mod-2 shadow is degenerate. Only the direct signed search decides.

8 Verification

Four independent layers check the 51 claim. (i) Every search certificate is replayed by a checker sharing no code with the search (chain validity, coverage, basis independence/unimodularity). (ii) The sign model is verified against the signed Brent equations. (iii) The generator that assembles the full scheme re-verifies the *emitted* program end-to-end over the word ring $\mathbb{Z}\langle A, B\rangle$ — all nine outputs equal AB as non-commutative polynomials, with the additions counted ($12 + 13 + 26 = 51$); this gate caught a genuine sign-bookkeeping bug on its first run. (iv) The Lean theorem of Appendix A is checked by the Lean 4 kernel against Mathlib's matrix product; the hypothesis-form theorem depends on `propext` alone, the Matrix-API form on Lean's three standard axioms, and the whole file re-elaborates in under ten seconds (the appendix header gives the exact command).

Stapleton and Perminov's scheme, kernel-checked. We applied the same pipeline to the *published* program of [15], transcribed verbatim from the verification listing in their paper (their reused variables SSA-renamed): the generator asserts their addition accounting (counted $13 + 12 + 26 = 51$, uncounted conversions $1 + 3 + 1 = 5$, one-shot 56), checks their P, Q, R basis matrices are unimodular (the amortization requirement), replays the program over $\mathbb{Z}\langle A, B\rangle$, and emits `proof/Mm/SP51.lean`, whose theorems carry the same axiom profile as ours. Both ring-valid 51-packages with known conversions are thus checked down to the Lean kernel. Their kernel's class-optimality is kernel-checked too: `proof/Mm/SP51Optimal.lean` proves that, from any nine starting vectors and after any invertible change of coordinates, chains of sums and differences that list the three legs of their scheme need at least 13, 12 and 12 additions — so $13+12+(12+14) = 51$ is optimal for the class, over $\mathbb{F}_2$ and over $\mathbb{Z}$ alike — by counting sum triples: each new vector a chain produces uses up a distinct triple of listed vectors adding to zero, and the three legs have only 9, 9 and 10; its finite facts are checked by `decide` in the kernel, on the three standard axioms (Appendix B sketches the statement and the proof).

Artifacts. All artifacts are in the repository https://github.com/gsidebottom/logic: the Lean proofs `proof/Mm/Ring51.lean`, `proof/Mm/SP51.lean` and `proof/Mm/SP51Optimal.lean`; the leg certificates, sign models and signed chains in `matmul/external/ks51/` and the

transcribed Stapleton–Perminov program and certificate in `matmul/external/sp51/`; the search tools (`src/bin/fbmin.rs`, `src/bin/fbsmin.rs`) and drivers (`matmul/fbsweep.py`, `matmul/fbsweep2.py`, `matmul/fbsloop.py`); and the generators `matmul/gen_ring51_lean.py` and `matmul/gen_sp51_lean.py`, whose word-ring gates re-verify the emitted programs on every run, and `matmul/gen_sp51_optimal_data.py`, which emits the optimality proof's data from their certificate. An archival deposit of these artifacts (this paper, the standalone proofs, the Lake project, certificates, tools, and a repository snapshot at tag `v2.0-ks51`) is published at doi:10.5281/zenodo.23305910, which resolves to its latest version.

9 Open problems

(1) Is the ring-valid optimum of `i19w225c4efh` 52 or 53? Equivalently: can a free-vector basis beat every 9-target basis over $\mathbb{Z}$? (2) Does any rank-23 class reach per-step 50, i.e. exact leg sum 36? The completed sweeps exclude it for the entire known database (minimum sum outside the four 51s: 38); a structural lower bound better than Proposition 3.2 would settle it in general. (3) Characterize the sign models under which a class's signed leg minima match its $\mathbb{F}_2$ minima — model 21's success after twenty failures (and orbit 15's for the fourth class) invites explanation, and `i12w219c23fg`'s 128 straight failures ask whether its 51 lifts at all. (4) Explain or refute the `i12` concentration of Section 6: does a rank-one-rich spectrum with a heavy slot *force* small legs? (5) The same machinery applies verbatim to 4×4 rank-47/48/49 schemes.

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A Lean formalization: our 51-addition scheme computes AB

The following is the complete scheme definition and the matrix-product theorem, verbatim from `proof/Mm/Ring51.lean` (comments lightly ASCII-folded for typesetting; the repository file is authoritative). Sections marked `UNCOUNTED` are the amortized basis changes; the `COUNTED` sections contain the $12+13+26 = 51$ per-step additions. To re-check the proof from the repository (forcing re-elaboration rather than a cached replay):

```
cd proof && find .lake/build -name "Ring51.*" -delete && lake build Mm.Ring51
```

Expected: `Built Mm.Ring51` in 6–8 seconds against the prebuilt Mathlib cache, with axiom audits `[propext]` (for the hypothesis-form theorem `Ring51.correct`, not reproduced here) and `[propext, Classical.choice, Quot.sound]` for the theorem below. The ambient context is `variable {R : Type _} [Ring R]`.

```
/-- The 51-addition-per-step scheme as a map on `3x3` matrices, in the
structure of `Matmul55.scheme`: the two UNCOUNTED unimodular input
basis changes, the COUNTED 12- and 13-addition input chains, the 23
products, the COUNTED 26-addition transposed output stage, and the
UNCOUNTED output basis change.  Counted additions per recursive step:
12 + 13 + 26 = **51**. -/
def rank23_add51_scheme (A B : Matrix (Fin 3) (Fin 3) R) :
    Matrix (Fin 3) (Fin 3) R :=
  -- A-side basis change (UNCOUNTED, amortized): the 9 transformed
  -- inputs, an invertible integer map applied once per matrix
  let ab0 := A 0 0
  let ab1 := A 1 0
  let ab2 := A 0 0 + A 2 0
  let ab3 := A 0 1 + A 2 1
  let ab4 := A 1 1
  let ab5 := A 2 1
  let ab6 := A 1 1 + A 1 2 - A 2 2
  let ab7 := A 2 2
  let ab8 := A 0 2
  -- A-side chain: 12 COUNTED additions (paid per recursive step)
  let aw0 := ab0 + ab5
  let aw1 := ab1 - ab7
```

```
let aw2 := ab2 - ab7
let aw3 := ab3 + ab8
let aw4 := ab4 - ab8
let aw5 := ab6 - aw1
let aw6 := aw0 - aw3
let aw7 := aw2 - aw3
let aw8 := aw4 - aw5
let aw9 := aw4 - aw6
let aw10 := ab1 - ab2
let aw11 := aw0 + aw10
-- B-side basis change (UNCOUNTED)
let bb0 := B 0 0
let bb1 := B 0 1 + B 1 1
let bb2 := B 0 2 + B 1 2
let bb3 := B 1 1
let bb4 := B 1 2
let bb5 := B 2 0 - B 2 2
let bb6 := B 1 1 - B 2 1
let bb7 := B 2 2
let bb8 := B 1 0
-- B-side chain: 13 COUNTED additions
let bw0 := bb0 - bb2
let bw1 := bb0 + bb7
let bw2 := bb4 + bw0
let bw3 := bb5 + bw1
let bw4 := bb6 + bb7
let bw5 := bb4 - bw4
let bw6 := bb5 + bw4
let bw7 := bb3 - bw6
let bw8 := bb8 + bw0
let bw9 := bb8 - bw6
let bw10 := bw1 - bw2
let bw11 := bb0 - bb1
let bw12 := bw6 + bw11
-- the 23 products (A factor on the left: recurses on blocks)
let m1 := aw0 * bw0
let m2 := aw1 * bw1
let m3 := aw2 * bb0
let m4 := ab0 * bb1
let m5 := ab1 * bw12
let m6 := ab2 * bw12
let m7 := ab0 * bb2
let m8 := ab1 * bw10
let m9 := aw11 * bw2
let m10 := ab3 * bw9
let m11 := ab4 * bw9
let m12 := ab5 * bw8
let m13 := aw6 * bb3
let m14 := aw4 * bw4
let m15 := aw3 * bw6
let m16 := aw9 * bb4
let m17 := ab4 * bw5
let m18 := ab6 * bb5
let m19 := ab7 * bw3
let m20 := ab8 * bb6
let m21 := aw5 * bw7
let m22 := aw7 * bw7
let m23 := aw8 * bb7
-- output stage: 26 COUNTED additions -- the Tellegen transpose
```

```
  -- of the certified 12-addition C chain (12 + (23-9) fan-in = 26)
  let z0 := m3 + m19
  let z1 := m14 + m20
  let z2 := z1 + m17
  let z3 := m23 - z2
  let z4 := m2 + m19
  let z5 := z4 - z3
  let z6 := m1 - z0
  let z7 := z6 + m7
  let z8 := m15 - m20
  let z9 := z8 + z0
  let z10 := m18 - m17
  let z11 := z10 + z5
  let z12 := m4 - m13
  let z13 := z12 - m20
  let z14 := m5 - m21
  let z15 := z14 - z11
  let z16 := m6 + m22
  let z17 := z16 - z9
  let z18 := m8 - z3
  let z19 := m9 - z5
  let z20 := z19 - z7
  let z21 := m10 + z9
  let z22 := m11 + z11
  let z23 := m12 - z7
  let z24 := m16 - z2
  let z25 := z24 + m7
  -- output basis change (UNCOUNTED): the inverse transform,
  -- rows of the unimodular output basis
  let c11 := z21 - z23
  let c12 := z13
  let c13 := z25
  let c21 := z22
  let c22 := -z15
  let c23 := z18
  let c31 := z23
  let c32 := -z13 - z17
  let c33 := z18 + z20
  !![c11, c12, c13;
     c21, c22, c23;
     c31, c32, c33]

-- `rank23_add51_scheme` unfolds to a ~120-binding straight-line
-- program; a single elaboration exceeds the default heartbeat budget.
set_option maxHeartbeats 1600000 in
-- Plain `simp` evaluates the concrete `!![..] i j` indexing via
-- Mathlib's matrix simprocs; the explicit lemma set is version-fragile,
-- so the flexible-`simp` lint is accepted here (as in Mm/Matrix.lean).
set_option linter.flexible false in
/-- The scheme computes the matrix product, in Mathlib's own terms. -/
theorem rank23_add51_scheme_eq_mul (A B : Matrix (Fin 3) (Fin 3) R) :
    rank23_add51_scheme A B = A * B := by
  simp only [rank23_add51_scheme]
  ext i j
  fin_cases i <;> fin_cases j <;>
    simp [Matrix.mul_apply, Fin.sum_univ_three] <;>
    noncomm_ring
```

B Lean formalization: Stapleton and Perminov's 51 is optimal for its class

The file `proof/Mm/SP51Optimal.lean` machine-checks the claim of Section 4 that the 51-addition kernel of Stapleton and Perminov [15] is exactly optimal for its class. This appendix sketches the statement and the proof; the repository file is authoritative. To re-check it:

```
cd proof && lake build Mm.SP51Optimal
```

Expected: `Built Mm.SP51Optimal` in about four minutes, nearly all of it kernel `decide`; both headline theorems depend only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`, with no `native_decide` and no `sorry`.

The statement. A linear form on 3×3 matrices over $\mathbb{F}_2$ is its coefficient matrix (`Form`). A *chain of n additions* from nine starting vectors is a list of $9+n$ vectors that begins with them, each later entry the sum or the difference of two earlier entries; a finite set is *computable with n additions* if some chain, from some nine starting vectors, lists all of it, and *computable after a change of basis* if the starting vectors can be taken linearly independent. Verbatim from the file (the ambient `G` is any additive commutative group):

```
def IsChain {n : ℕ} (s : Fin 9 → G) (v : Fin (9 + n) → G) : Prop :=
  (∀ i, v (Fin.castAdd n i) = s i) ∧
    ∀ k : Fin n, ∃ i j : Fin (9 + n), i < Fin.natAdd 9 k ∧ j < Fin.natAdd 9 k ∧
      (v (Fin.natAdd 9 k) = v i + v j ∨ v (Fin.natAdd 9 k) = v i - v j)

def Computable (T : Finset G) (n : ℕ) : Prop :=
  ∃ (s : Fin 9 → G) (v : Fin (9 + n) → G), IsChain s v ∧ ∀ t ∈ T, ∃ i, v i = t

def ComputableFromBasis (T : Finset Form) (n : ℕ) : Prop :=
  ∃ (s : Fin 9 → Form) (v : Fin (9 + n) → Form),
    LinearIndependent (ZMod 2) s ∧ IsChain s v ∧ ∀ t ∈ T, ∃ i, v i = t

theorem kernel_optimal_for_class (eA eB eC : Form ≃ₗ[ZMod 2] Form) :
    (∀ nA nB nC : ℕ, Computable (legA.image eA) nA → Computable (legB.image eB) nB →
        Computable (legC.image eC) nC → 51 ≤ nA + nB + (nC + 14)) ∧
      ComputableFromBasis (legA.image eA) 13 ∧ ComputableFromBasis (legB.image eB) 12 ∧
      ComputableFromBasis (legC.image eC) 12
```

Here `legA`, `legB`, `legC` are the legs of their scheme, written out in the file as literal matrices: the 20 distinct nonzero rows of U and of V and columns of W of their certificate, mod 2. The maps `eA`, `eB`, `eC` range over all invertible linear changes of coordinates. The symmetries of matrix multiplication act on each side of a scheme by such a map (and may permute the sides), so every scheme in the class presents its legs in this form, and the per-step cost $a_A + a_B + (a_C + 14)$ is symmetric in the three legs. The lower bounds hold from *any* nine starting vectors, a basis or not; the upper bounds start from a basis, as the Karstadt–Schwartz model requires. A companion theorem, `kernel_optimal_over_int`, gives the same lower bound for chains over $\mathbb{Z}$ whose entries reduce mod 2 to the legs. Two ingredients come from the cost model rather than from the formalization: the output side's $+14$ (the transposition principle) and the description of the class action just given.

The counting lemma. For a chain whose nonzero entries form the set S, call a three-element subset of S that sums to zero a *sum triple*, and let $\tau(S)$ be their number. An entry $c \in S$ that is

not a starting vector is first listed as $c = a \pm b$ with a and b listed earlier; over $\mathbb{F}_2$ this makes a and b nonzero, distinct and different from c, so $\{a, b, c\}$ is a sum triple, and $c \mapsto \{a, b, c\}$ is injective because a triple yields only its last-listed member. Hence

$$|S| \le 9 + \tau(S),$$

and $|S|+1 \le 9+\tau(S)$ whenever some nonempty family K of sum triples covers each of its members at least twice: the last-listed member of $\bigcup K$ completes two triples of K at once, so one triple of K produces nothing (`card_vals_le`).

The three legs. Each leg has 20 forms but few sum triples:

leg	forms d	sum triples τ	floor $d-9$	exact cost
A (rows of U)	20	9	11	13
B (rows of V)	20	9	11	12
C (columns of W)	20	10	11	12

A chain of at most 11 additions lists at most 20 vectors, hence exactly the leg, and the lemma would need $20 \le 9+\tau$, that is $\tau \ge 11$; so every leg needs at least 12 additions, which settles B and C. For A with 12 additions, the chain lists the leg and at most one further nonzero vector h, and $21 \le 9+\tau(A\cup\{h\})$. A sum triple through h has the form $\{h, a, a+h\}$, so $\tau(A\cup\{h\}) \le 9+\lfloor m_h/2 \rfloor$ with $m_h = \#\{a \in A : a+h \in A\}$, which forces $m_h \ge 6$. Exactly five vectors have $m_h \ge 6$, and for each of them $\tau(A \cup \{h\}) = 12$ and a nonempty family of its triples covers each of its members at least twice, so the lemma gives $22 \le 21$. So leg A needs at least 13 additions.

What the kernel checks. The finite facts — each leg's size, the absence of zero, and its triple count; the five vectors (a check over all 512 forms) and their families; and three optimal chains with linearly independent starting vectors, of 13, 12 and 12 additions — are decided by `decide` in the kernel, from data emitted from their certificate by `matmul/gen_sp51_optimal_data.py`. Invertible linear maps carry chains to chains, which gives the class statement; reduction mod 2 carries integer chains to chains over $\mathbb{F}_2$, which gives `kernel_optimal_over_int`.